

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0020

Hemet 345 kV
June 28, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0020. The Interconnection Customer proposes a 919.3 MW natural gas combined cycle facility interconnecting at the Hemet 345 kV substation in the Pennsylvania area. The request was received on March 16, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 17 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$144,501,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Dekalb-Robison Park 230kV line
Contingency: Jones Chapel-Dekalb 500kV
Loading: 110.2% (Rating: 777 MVA)
DFAX: 6.0%
Pre-project loading: 103.1%
2. Facility: Hemet 138/1 kV Autotransformer
Contingency: Hemet-Filer 345kV Line
Loading: 104.9% (Rating: 513 MVA)
DFAX: 55.3%
Pre-project loading: 5.8%
3. Facility: Robison Park-Russell Fork 230kV
Contingency: Dekalb-Jones Chapel 500kV
Loading: 103.6% (Rating: 771 MVA)
DFAX: 7.8%
Pre-project loading: 94.3%
4. Facility: West Campus-Ft Humbug 345kV
Contingency: Orchard Rd-Fort Humbug 500kV line
Loading: 102.2% (Rating: 1,166 MVA)
DFAX: 8.2%
Pre-project loading: 95.8%
5. Facility: Russell Fork-Bluebonnet 230kV
Contingency: Jones Chapel-Dekalb 500kV Line
Loading: 102.0% (Rating: 607 MVA)
DFAX: 5.7%
Pre-project loading: 93.4%

6. Facility: Concord-Bluebonnet 230kV
Contingency: Jones Chapel-Dekalb 500kV
Loading: 100.8% (Rating: 679 MVA)
DFAX: 5.2%
Pre-project loading: 93.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dekalb-Robison Park 230kV line	Rebuild line with higher-capacity conductor	\$48,114,000	100% (\$48,114,000)
Hemet 138/1 kV Autotransformer	Replace transformer with higher-rated unit	\$4,372,000	100% (\$4,372,000)
Robison Park-Russell Fork 230kV line	Reconductor line	\$17,705,000	100% (\$17,705,000)
W Campus-Ft Humbug 345kV L	Reconductor line	\$48,189,000	100% (\$48,189,000)
Bluebonnet-Russell Fork 230kV	Reconductor line	\$19,987,000	100% (\$19,987,000)
Bluebonnet-Concord 230kV line	Reconductor line	\$6,134,000	100% (\$6,134,000)
	Total allocated to project		\$144,501,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.